

Chapter 4 Software Requirement Specifications

In this section a detailed description of all related software requirements for our project is presented. Our project comprises a listing of the characteristics, functional specifications, quality standards, non-functional requirements, related assumption, hardware and software requirement, scenarios, user interface design and analysis of the risks associated with our project.

4.1 List of Features

The following features will be available in the system.

- Healthcare providers should have secure sign-up and log-in functionality.
- Compatibility for uploading different brain scan datasets like MRI images.
- Pre-processing of brain scans.
- Various trained datasets based on different models for differentiating brain tumor or stroke from hemorrhage.
- Display suggestions if the scan has any brain condition.

4.2 Functional Requirements

The following functional requirements shall be available in the system.

- The user shall be able to create an account by filling out the information required.
- The user shall be able to login to their account by providing credentials and use the software.
- The user shall be able to logout of their account.
- The user shall be able to upload their brain scans in multiple formats.
- The system shall convert the images into the required format.
- The system shall implement a machine learning algorithm to classify whether the scan is cancerous or not.
- The system shall implement machine learning algorithms for the image classification of various brain conditions (stroke, Alzheimer, and hemorrhage).
- The system shall provide a diagnostic result.
- The system shall display recommendations to the user based on their brain condition.

4.3 Quality Attributes

The project aims to incorporate the following fundamental quality attributes in order to ensure the prevention of application defects. Each attribute shall be used to evaluate the application quality and performance during operation.

4.3.1 Reliability

It is important to ensure the reliability of the project as it is designed for medical purposes. We need to make sure that the application gives the right results. It does not matter where it is used or what conditions it is working in. The product should always be accurate and trustworthy. This is important because people rely on the app to provide correct information in medical situations and we want to make sure it does that every time.

4.3.2 Maintainability

SynapSure ensures long term viability with a set of pre-trained segmentation models for brain scans analysis. While the current algorithms represent the latest advancements the system is designed to seamlessly integrate future algorithms which ensures continuous state-of-the-art diagnostic capabilities. Considering this need the application shall have enough maintainability to update the algorithms or switch the underlying technologies for efficient inferences later on.

4.3.3 Usability

The project aims to serve as a utility for general public and medical professionals to assist in their diagnostic workflow. The project must have high usability and accessibility as the audience is non technical. The usability of the project shall be evaluated with the following performance indicators.

- Consistent user interface across the application to avoid any confusion.
- Easy for new or infrequent users to learn to use the system.
- Easy for users to input data and interpret the output.

4.3.4 Correctness

Considering a medical context, it is critical that the application produces correct results and avoids false positives. In order to achieve this goal the project is committed to following the functional requirements closely. The focus is on preventing false positives which means the system should not incorrectly identify something as true when it is not true.

4.4 Non-Functional Requirements

4.4.1 Availability

- The system shall be readily available to give response to the user's requests.
- The system should be available to always detect brain diseases 24/7.

4.4.2 Reusability

- The code should be divided into modules based on the functionality so that it favors code reusability.
- The system should be able to classify brain scans on a consistent basis and should not fail after a specific period of working capability.

4.4.3 Robustness

- The system shall able to process and deal with extensive amounts of data without loss of critical information and operational capability.
- The system shall preserve the progress, in case of connection dropouts or hardware failures.

4.4.4 Security Requirements

- The system shall allow the users to have a unique email/phone number of their accounts from which the only particular user can access his account.
- The system shall not be able to share the user data with anyone.

4.4.5 Performance

- The system shall be able to give responses to the user's request in real time.

4.5 Assumptions

The assumptions made for the specification for the user to use the system are listed as:

- The user has sound knowledge of the project's purpose and utility.
- The user will input their brain scan. It is assumed that the users will have their brain scans.
- The users have basic knowledge of how to use a computer and website.

4.6 Use Cases

Following are the use cases identified in our web application.

4.6.1 Sign Up

Name	Sign Up												
Actors	User												
Summary	Users create a new account on the SynapSure platform by providing required registration information.												
Pre-Conditions	The user must fill out the whole sign up form. None of the user’s information exists in the database.												
Post-Conditions	Account is successfully created. The user is sent to the login page.												
Special Requirements	Valid email address and secure password												
Basic Flow													
<table><tr><th colspan="2">Actor Action</th><th colspan="2">System Response</th></tr><tr><td>1</td><td>User selects the 'Sign Up' option.</td><td>2</td><td>System displays the registration form.</td></tr><tr><td>3</td><td>User provides the required information, including email and password.</td><td>4</td><td>System validates the information and create account.</td></tr></table>		Actor Action		System Response		1	User selects the 'Sign Up' option.	2	System displays the registration form.	3	User provides the required information, including email and password.	4	System validates the information and create account.
Actor Action		System Response											
1	User selects the 'Sign Up' option.	2	System displays the registration form.										
3	User provides the required information, including email and password.	4	System validates the information and create account.										
Alternative Flow													
3	User enters incorrect input fields.	4-A	System displays an error message: 'Invalid Information Entered'.										
3	Any necessary input fields are omitted by the user.	4-B	System displays an error message: 'Please fill in the missing field(s)'.										

4.6.2 Login

Name		Login	
Actors		User	
Summary		Users access their accounts by providing their login credentials.	
Pre-Conditions		User must have a registered account on SynapSure. User has a valid email and password.	
Post-Conditions		User is successfully logged into the account. The user is redirected to the SynapSure dashboard.	
Special Requirements		Valid email address and password	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Login' option.	2	System displays the login screen which requests email and password.
3	User inputs a valid password and email.	4	System authenticates the user's email and password and redirects the user to SynapSure's dashboard.
Alternative Flows			
3	User enters incorrect login credentials.	4-A	System displays an error message: 'Invalid Login Credentials'.

4.6.3 Upload Brain Scan

Name	Upload Brain Scan		
Actors	User		
Summary	Users upload brain scan images into the SynapSure system. The system then prepares the scans for analysis.		
Pre-Conditions	User is registered and logged into the system. The brain scan image is in a supported format.		
Post-Conditions	The brain scan is successfully uploaded and queued for analysis.		
Special Requirements	High-resolution scan images		
Basic Flow			
Actor Action		System Response	
1	User selects the 'Upload Scan' option.	2	System displays the upload interface.
3	User uploads the brain scan image.	4	System confirms the upload and queues the scan for analysis.
Alternative Flow			
3	Medical professional attempts to upload an unsupported file format.	4-A	System displays an error message: 'Unsupported File Format'.

4.6.4 View Analysis Result

Name		View Analysis Result	
Actors		User	
Summary		Users access and review the analysis results of uploaded brain scans.	
Pre-Conditions		User is logged into the system. A brain scan has been analyzed by the system.	
Post-Conditions		User has viewed and understood the analysis results.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User selects the 'View Results' option for a specific scan.	2	System displays the analysis results.
Alternative Flow			
1	Medical professional selects a scan that is still being analyzed.	2-A	System displays a message: 'Analysis still in progress. Please check back later.'

4.6.5 Provide Feedback

Name	Provide Feedback
Actors	User
Summary	Users provide feedback on their experience with the system includes suggestions for improvements, reporting issues and general comments.
Pre-Conditions	User is registered and logged into the system.
Post-Conditions	Feedback is recorded and available for system evaluation.

Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Provide Feedback' option.	2	System displays the feedback form.
3	User fills out the feedback form and submits it.	4	System confirms receipt of the feedback.
Alternative Flow			
3	User submits an incomplete form.	4-A	System prompts the user to complete all required fields.

4.6.6 Generate Report

Name		Generate Report	
Actors		User	
Summary		After analyzing a brain scan, users can generate a detailed report that includes the diagnosis and other relevant findings.	
Pre-Conditions		User is logged in. A brain scan has been analyzed and results are available.	
Post-Conditions		A detailed diagnostic report is generated and available for review or download.	
Special Requirements		Report must be clear, concise, and medically accurate.	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Generate Report' option for a specific scan.	2	System compiles the analysis results into a comprehensive report format.
3	User reviews or downloads the generated report.	4	System provides options to review on-screen or download the report in a preferred format.
Alternative Flow			
3	An error occurs during report generation.	4-A	System displays an error message and suggests retrying or contacting support.

4.7 Hardware and Software Requirements

The project employs a multitude of hardware and software services for its operations. For the purpose of developing and deploying this project following hardware and software requirements need to be fulfilled.

4.7.1 Hardware Requirements

The following are the hardware requirements required for the operation of the system:

- A computer system will be required to develop this website. Core i5 or later processors will be ideal for development of this web-based project.
- The device shall have the sufficient space available to store the brain scans.
- Internet connection is required to set up the system.

4.7.2 Software Requirements

The following are the software requirements required for the operation of the system:

- Anaconda
- Python, 3.8 or above
- Web Browser i.e. Google Chrome or Firefox
- MongoDB
- Python Libraries
- GitHub
- React
- JavaScript
- Draw.io

4.8 Graphical User Interface

This section should give the GUI dumps of each screen, with reference to the users. The navigation flow of each user is also required, and each GUI should mark the functionality/use case that it covers.

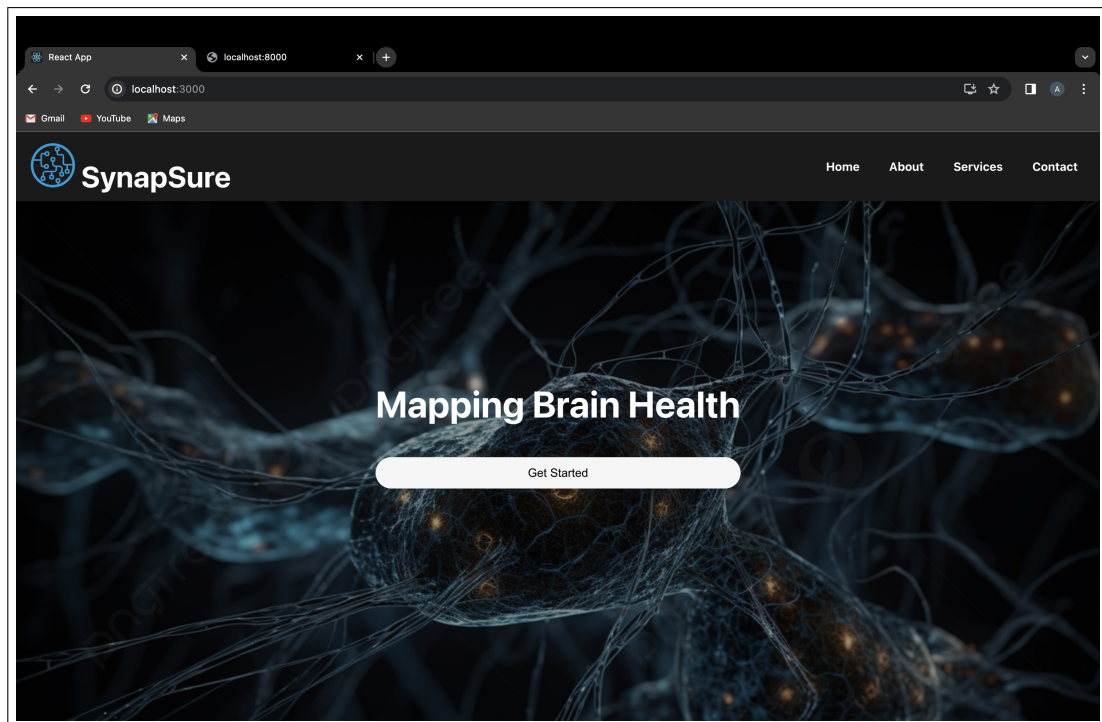


Figure 4.1: Landing Page

Landing page shows a navbar with options: Home, About, Services, and contact. Also, there is a button at the center of the page showing the option of "Get Started"

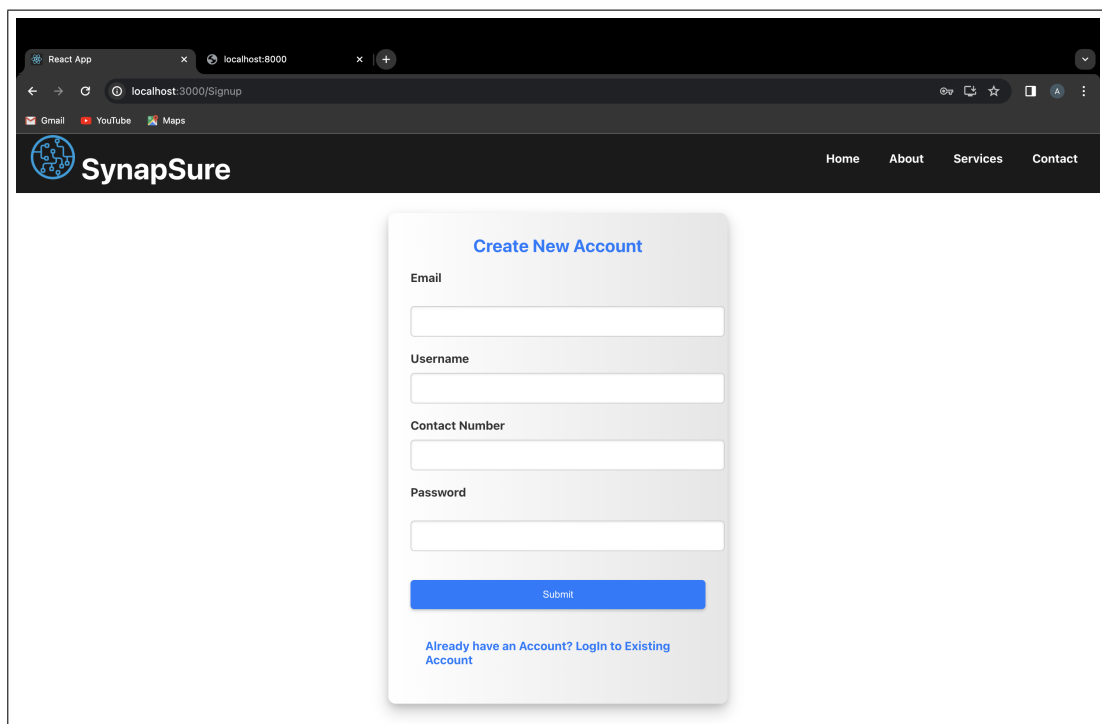


Figure 4.2: Sign-up Page

User friendly Sign-up Page for quick and easy registration. It includes the Email, Username, contact number and the password. There is also an option for already existing account

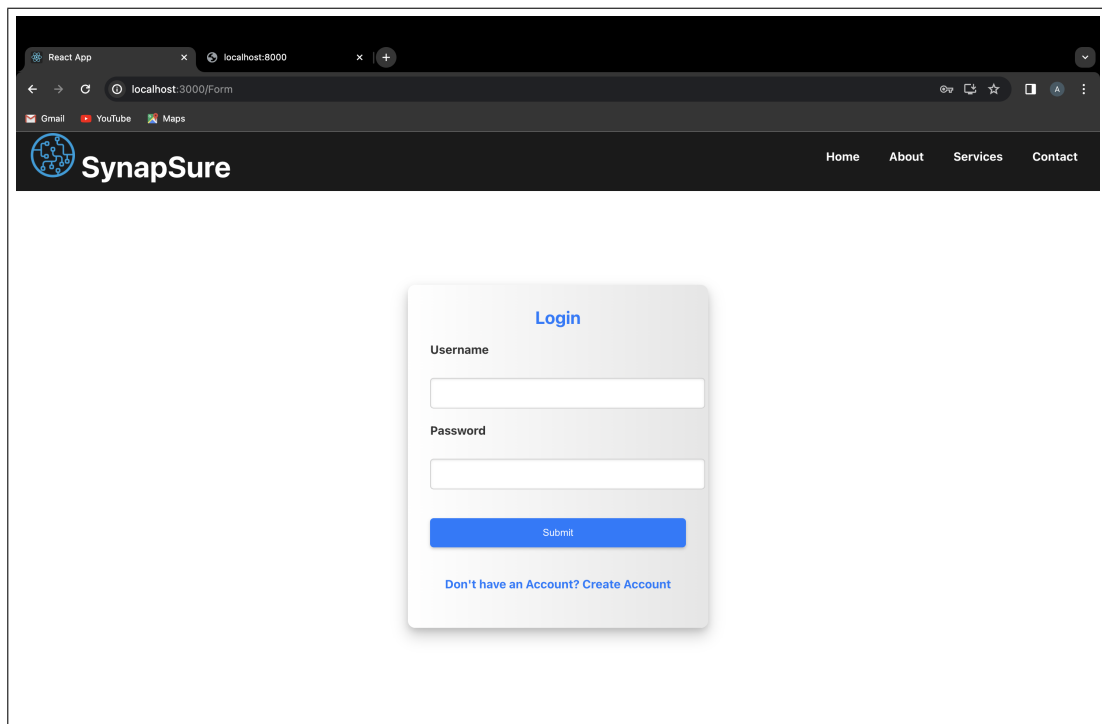


Figure 4.3: Login Page

Login Page for secure and easy access to SynapSure's dashboard. This figure includes the Username and password and there is also an option for creating a new account.

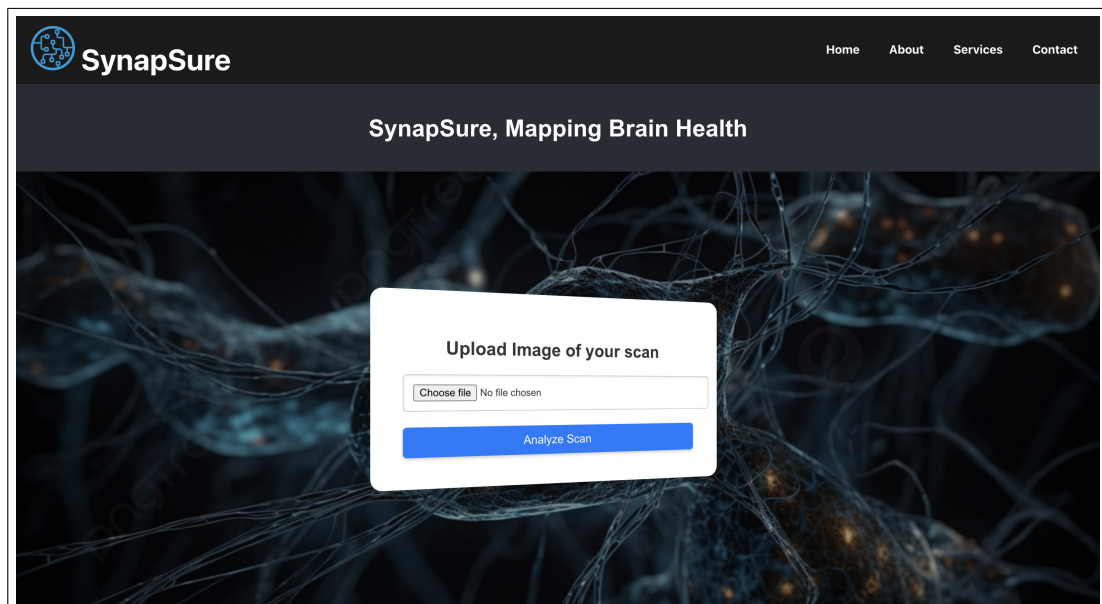


Figure 4.4: Upload Scan Page

Page for uploading brain scans allow user to upload their scan for analysis. This figure shows that the web page has a choose file button and then a button for analyzing the scan

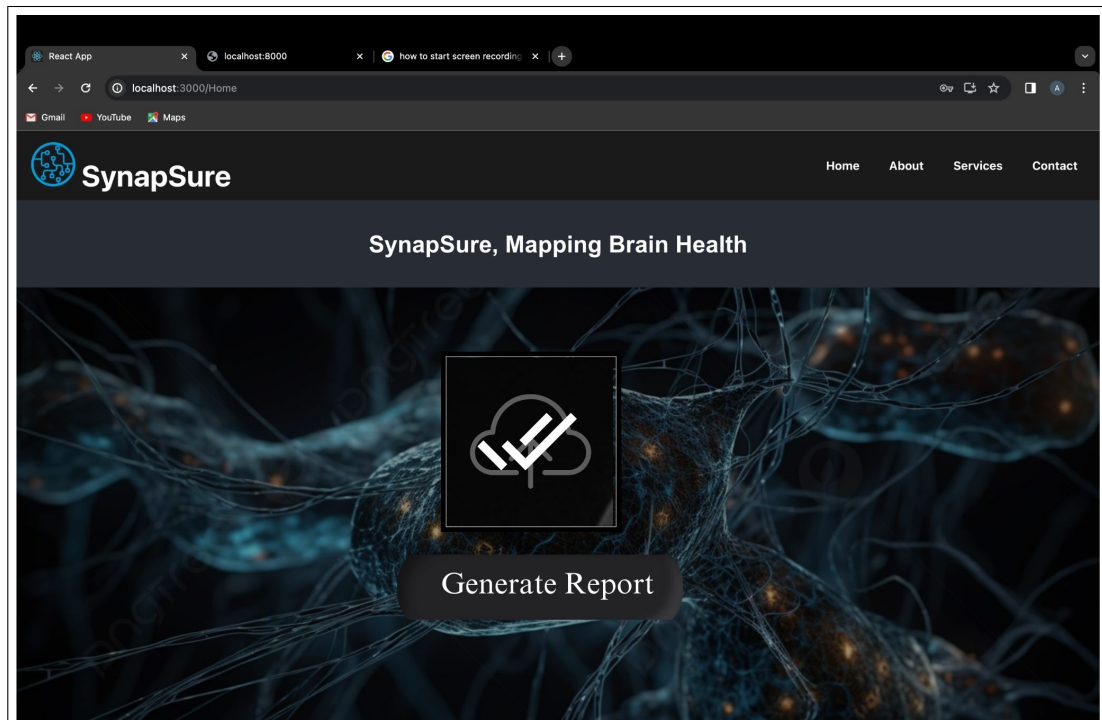


Figure 4.5: Generate Report Page

Page that allow user to generate analysis report by clicking on "Generate Report" button

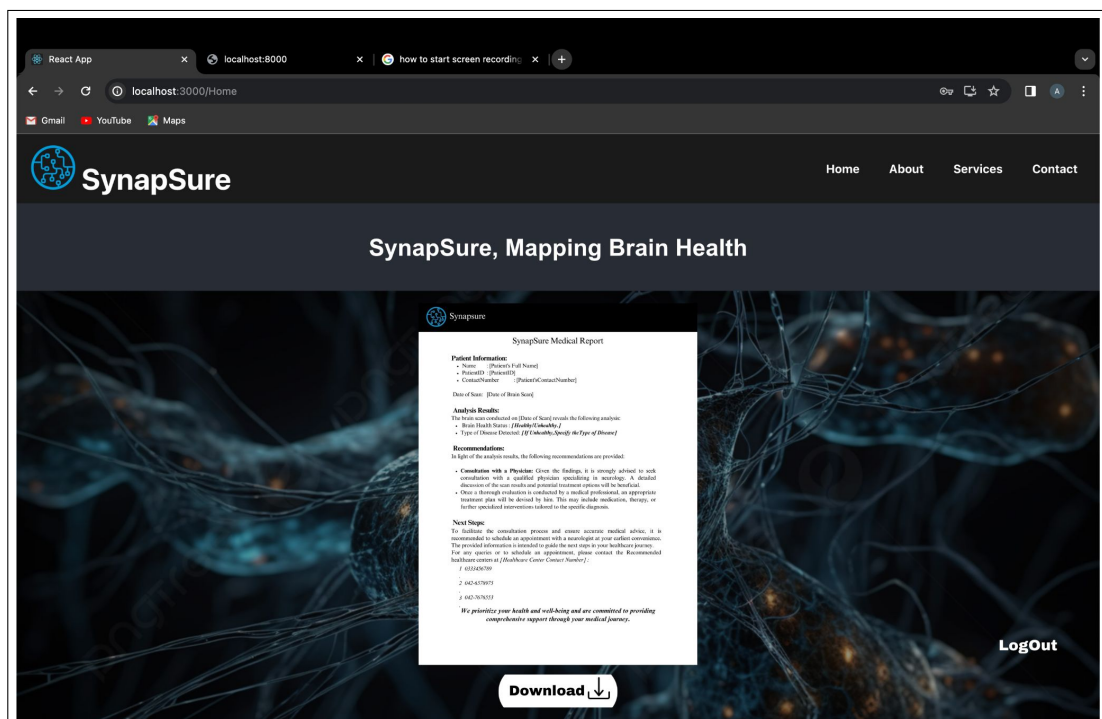


Figure 4.6: Report Page

Page displaying detailed diagnostic report and the option to download the report. It gives all the details of the patient and all details regarding the condition of brain



Synapsure

SynapSure Medical Report

Patient Information:

- Name : [Patient's Full Name]
- PatientID : [PatientID]
- ContactNumber : [Patient'sContactNumber]

Date of Scan: [Date of Brain Scan]

Analysis Results:

The brain scan conducted on [Date of Scan] reveals the following analysis:

- Brain Health Status : *[Healthy/Unhealthy.]*
- Type of Disease Detected: *[If Unhealthy,Specify theType of Disease]*

Recommendations:

In light of the analysis results, the following recommendations are provided:

- **Consultation with a Physician:** Given the findings, it is strongly advised to seek consultation with a qualified physician specializing in neurology. A detailed discussion of the scan results and potential treatment options will be beneficial.
- Once a thorough evaluation is conducted by a medical professional, an appropriate treatment plan will be devised by him. This may include medication, therapy, or further specialized interventions tailored to the specific diagnosis.

Next Steps:

To facilitate the consultation process and ensure accurate medical advice, it is recommended to schedule an appointment with a neurologist at your earliest convenience. The provided information is intended to guide the next steps in your healthcare journey. For any queries or to schedule an appointment, please contact the Recommended healthcare centers at *[Healthcare Center Contact Number]* :

- 1 0333456789
- .
- 2 042-6578975
- .
- 3 042-7676553
- .

We prioritize your health and well-being and are committed to providing comprehensive support through your medical journey.

Figure 4.7: Medical Report

Sample diagnostic report offering insights of the analysis results of brain scans performed by SynapSure. It includes the patient information which includes name and personal information. There is analysis results and recommendations and last is the next steps.

4.9 Database Design

4.9.1 ER Diagram

The relation of entities User, login, sign up and upload scan are shown with their relationship in crow foot notation such as a user can have one and only one account, a user can select zero or many scans to upload as shown in the figure below.

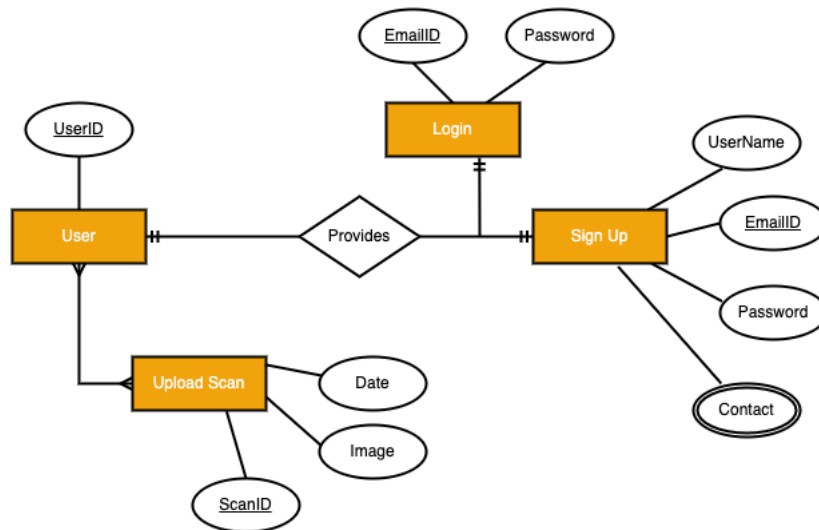


Figure 4.8: ER Diagram of SynapSure

Structure of the SynapSure project, the connections and dependencies between various data entities. It shows attributes for all

4.9.2 Data Dictionary

The data dictionary of the database, which will be used by the system, is shown below in the table.

Table 4.3: Data Dictionary

Overview of entities, attributes, data types, and descriptions in the database. There are five columns in this table

Entity	Attribute	Data Type	Nullable	Description
User	userID	Integer	No	The unique number assigned to the user.
Login	emailId password	String String	No No	The registered email address of the users. Password to authorize users.
Sign up	userName emailId password contact	String String String String	No No No Yes	User's full name. Unique email address of the user to register them in the system. Password to authorize users. The user's contact number which he can use for communication.
Upload Scan	scanId image date	int Image Date	No No No	The unique id of the image scan Image of the brain scan The date on which the brain was scanned.

4.10 Risk Analysis

As SynapSure aims for the diagnosis of brain scans it is crucial to conduct a comprehensive risk analysis to identify potential challenges and devise effective mitigation strategies. By anticipating and addressing these risks SynapSure can position itself for a successful and sustainable tool in brain diseases classification.

4.10.1 Technical Risks

The main technical risk that is considered concerns the availability and quality of brain scan data sets. For SynapSure to succeed there must be a wide array of large data sets for training and validation of machine learning algorithms. Biased or low sample data sets could lead to poor performing algorithms that will affect the precision with which brain conditions are classified. Further, with the advent of new machine learning and medical imaging technologies it is also possible that SynapSure's algorithms will become obsolete. It is necessary to always keep the latest methods of treatment as well as include them to SynapSure so that it remains significant and competitive.

4.10.2 Business Risks

SynapSure is an entity that operates in healthcare technology and in this domain, the rules, the market and competition dynamics change very fast. Changes on health policies or privacy rules might be introduced into the market of the project or other different diagnostic alternatives. Other emerging competitors such as other diagnostic techniques are significant business risks that require a good positioning and ongoing innovation in order to be ahead of them. Success of the project depends on effective cooperation with healthcare centers and any disturbances in this union could negatively influence the adoption and implementation of SynapSure.

4.10.3 Ethical and Privacy Risks

Since the medical information is very sensitive ethical and privacy issues dominate. It is important to the project that patients data be protected and they comply with privacy regulations. Breakdowns in data security will result to legal penalties degrade project image and lose trusts of users. To address ethical and privacy risks data protection techniques access controls and transparent data handling practices have to be implemented.

4.11 Conclusion

In this chapter, the software requirements for SynapSure are thoroughly discussed. The chapter provides a comprehensive road map for development by encompassing features, functional and non-functional requirements, quality attributes, and a graphical user interface. It emphasizes the need for reliability, maintainability, usability, and correctness which ensures the application's accuracy in medical contexts. Additionally, essential hardware and software requirements, user assumptions and comprehensive risk analyses addressing technical, business, and ethical concerns are also explained in the chapter. This information is like a guide for the next steps in making SynapSure a good and trustworthy tool for understanding brain diseases.

Chapter 5 Proposed Approach and Methodology

The SynapSure project adopts a modular approach consisting of Data Input, Preprocessing, Regression and Classification modules to achieve its goal of accurately diagnosing brain health conditions. This chapter provides detailed insights into each modules functionalities algorithms and significance within the system.

5.1 Data Input

The system initiates by acquiring diverse MRI brain scan data sets encompassing images and raw data from users. Leveraging the EfficientNet model the project reads and processes these scans. EfficientNet is selected for its efficient feature extraction capabilities crucial for accurate classification. These approaches ensures compatibility with multiple image format and seamless integration with subsequent module.

5.2 Data Preprocessing

Raw MRI data often contains noise that can hinder accurate classification. The Data Preprocessing module addresses this challenge using techniques like image normalization, filtering and enhancement. This module employs standardization of image sizes noise reduction through filters and enhancement of relevant features. Traditional image processing and deep learning methodologies are combined to optimize data quality.

5.3 Regression

To differentiate between healthy and unhealthy brain scans the Regression module is introduced. Utilizing regression analysis the system assesses quantitative indicators within the scans and establishing a baseline for health. Regression helps identify patterns and deviations associated with unhealthy conditions which establishes the groundwork for subsequent classification. The regression model is trained on a diverse data set to capture a broad spectrum of health variations.

5.4 Classification

Following regression analysis the project utilizes the EfficientNet model for further classification of unhealthy brain scans. The model pre trained on extensive data sets ensures adaptability and accuracy in identifying specific conditions like tumors, strokes and Alzheimer's. Transfer learning is employed to